

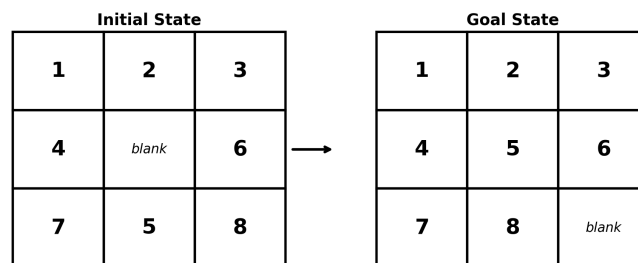
Daffodil International University
Faculty of Science & Information Technology
Department of Computer Science & Engineering
Quiz Examination

Course Title: Artificial Intelligence	Course Code: CSE 411
Time: 45 Minutes	Total Marks: 15
Date: 24-06-2026	

Instruction: Answer all questions. Do not use Alpha-Beta Pruning.

Question 1: 8-Puzzle Search Problem [5 Marks]

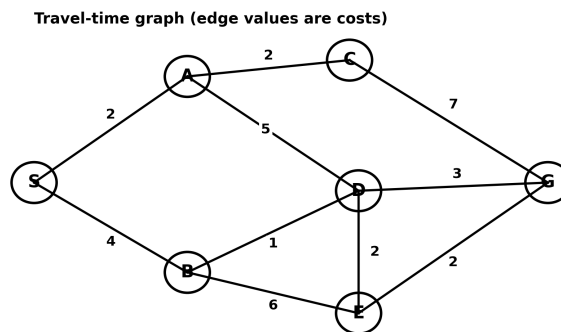
A robot tile system uses an 8-puzzle board. The blank tile can move Left, Right, Up, or Down when the move is valid. Consider the following initial and goal states:



Formulate the above 8-puzzle as a search problem and find a possible sequence of moves from the initial state to the goal state. Your answer should include state representation, initial state, goal test, possible actions of the blank tile, path cost assumption, and final move sequence.

Question 2: A* Search Using a Realistic Road Graph [5 Marks]

A rescue vehicle needs to travel from Campus Gate (S) to Hospital (G). Each edge value in the graph represents travel time in minutes. The heuristic $h(n)$ estimates the remaining travel time to the hospital.

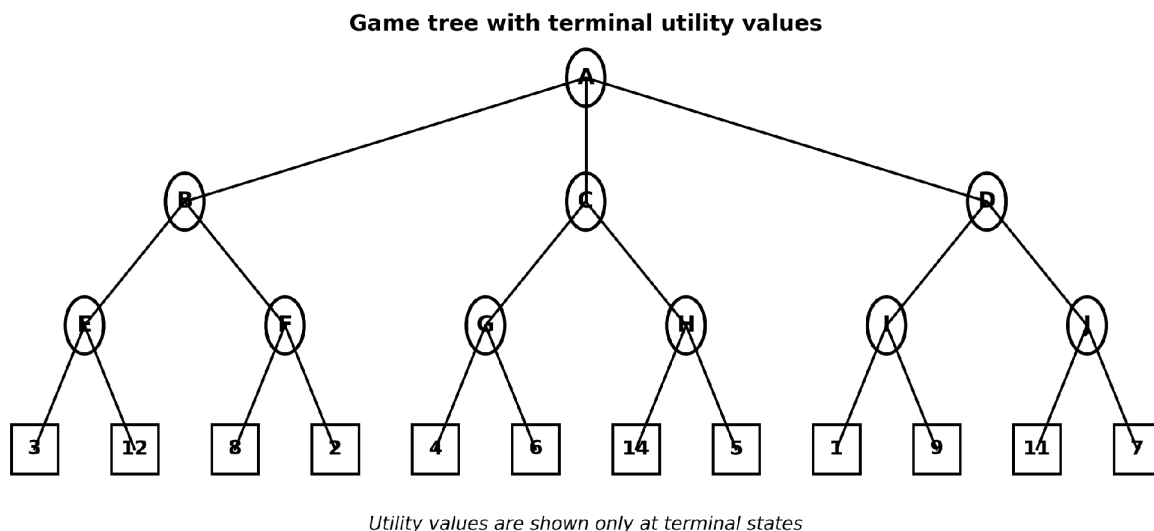


Node	S	A	B	C	D	E	G
$h(n)$	7	6	4	5	2	1	0

Using A* Search, find the best path from S to G. Show the calculation of $f(n) = g(n) + h(n)$ for the expanded/selected nodes and write the final path with total cost.

Question 3: Minimax Game Tree [5 Marks]

A computer is playing a two-player game. The computer makes the first move from node A. After that, the players alternate turns until a terminal state is reached. The terminal utility values are shown from the computer's point of view.



Using the Minimax algorithm, calculate the values of the non-terminal nodes and determine which first move the computer should choose. Students must identify the maximizing and minimizing levels by themselves from the turn order. Do not use Alpha-Beta Pruning.